

BETATAB 100MG TABLET

DESCRIPTION

A white to off white, round, biconvex tablet with scoreline on one side.

COMPOSITION

Each tablet contains Metoprolol Tartrate 100mg.

PHARMACODYNAMICS

Metoprolol tartrate is a cardio-selective beta-adrenergic blocking agent. It has a relatively greater blocking effect on beta₁-receptors (ie those mediating adrenergic stimulation of heart rate and contractility and release of free fatty acids from fat stores) than on beta₂-receptors, which are chiefly involved in broncho and vasodilation.

PHARMACOKINETICS

Absorption

Metoprolol is readily and completely absorbed from the gastrointestinal tract. Metoprolol is absorbed fully after oral administration. Within the therapeutic dosage range, the plasma concentrations increase in a linear manner in relation to dosage. Peak plasma levels are achieved after approx. 1.5–2 hours. Even though the plasma profile displays a broader interindividual variability, this appears to be easily reproducible on an individual basis. Due to the extensive first-pass effect, bioavailability after a single oral dose is approx. 50%. After repeated administration, the systemic availability of the dose increases to approx. 70%. After oral intake with food, the systemic availability of an oral dose increases by [SIC] approx. 30–40%.

Distribution

Peak plasma concentrations occur about 1½ hours after a single oral dose. Peak plasma metoprolol concentrations at steady state with usual doses have been reported as 20-340ng/ ml. Metoprolol is widely distributed, it crosses the bloodbrain barrier, the placenta. It is slightly bound to plasma protein.

Biotransformation

Metoprolol is metabolised through oxidation in the liver mainly by the CYP2D6 isoenzyme. Even though three main metabolites have been identified, none of them has a clinically significant beta-blocking effect. Generally, 95% of an oral dose is found in the urine. Only 5% of the dose is excreted unmodified via the kidneys; in isolated cases, this figure can reach as high as 30%. The elimination half-life of metoprolol averages 3.5 hours (with extremes of 1 and 9 hours). Total clearance is approx. 1 litre/minute. It is extensively metabolised in the liver; O-dealkylation followed by oxidation and aliphatic hydroxylation. The rate of hydroxylation to alpha-hydroxymetoprolol is reported to be determined by genetic polymorphism; the half-life of metoprolol in fast hydroxylators is stated to be 3-4 hours, whereas in poor hydroxylators it is about 7 hours.

Elimination

The metabolites are excreted in the urine together with only small amounts of unchanged metoprolol. Metoprolol is excreted in breast milk.

Special Population

Elderly: In comparison with administration to younger patients, the pharmacokinetics of metoprolol when administered to older patients shows no significant differences.

Renal impairment:

Renal dysfunction has barely any effect on the bioavailability of metoprolol. However, the excretion of metabolites is reduced. In patients with a glomerular filtration rate of less than 5 ml/minute, a significant accumulation of metabolites has been observed. This accumulation of metabolites, however, produces no increase in the beta blockade.

Hepatic impairment:

The pharmacokinetics of metoprolol are influenced only minimally by reduced hepatic function. However, in patients with serious hepatic cirrhosis and a portacaval shunt, the bioavailability of metoprolol can increase, and the total clearance can be reduced. Patients with portacaval anastomosis had a total clearance of approx. 0.3 litres/minute and AUC values that were 6 times higher than those found in healthy persons.

Severe angina pectoris:

Intrinsic sympathomimetic activity (ISA) may be a disadvantage for the patient with severe angina pectoris. There are however no indications that the efficacy in hypertensives is influenced by this characteristic. In exceptional cases, however, very high dosages can cause the ISA to predominate over the beta-adrenergic blocking capacity so that restriction of the maximum dosage is indicated.

Respiratory impairment:

It has not been proven that beta-blockers with ISA give a lower risk for bronchospasm or enhancement of preexisting bronchospastic complaints.

INDICATION

Hypertension: to reduce blood pressure and to reduce the risk of cardiovascular and coronary mortality (including sudden death), and morbidity.

Angina pectoris.

Disturbances of cardiac rhythm including especially supraventricular tachycardia.

Confirmed or suspected myocardial infarction - prevention of cardiac death and reinfarction.

Functional heart disorders with palpitations.

Migraine prophylaxis.

Hyperthyroidism.

RECOMMENDED DOSAGE

Dosage should be adjusted individually to avoid bradycardia. The tablets should be taken on an empty stomach. Concomitant intake of food increases the bioavailability of metoprolol with 40%. The following is valid as guidelines:

Hypertension

100-200 mg daily divided on one or two occasions. The tablets should be taken in the morning if once-daily dosage. In patients not responding to 200 mg, the dose could be combined with other antihypertensive agents, preferably diuretics and calcium antagonists of the dihydropyridine type, or increased.

Angina Pectoris

100-200 mg daily divided on two occasions. If needed, the dose can be combined with nitrates or increased.

Cardiac Arrhythmias

100-200 mg daily divided on two occasions. If needed, other antiarrhythmic agents may be added.

Prophylactic Therapy After Myocardial Infarction

As maintenance dosage: 100 mg morning and evening.

Functional Heart Disorders with Palpitations

100 mg once daily, given as a single dose in the morning. If needed, the dose can be increased to 200 mg

Migraine Prophylaxis

100-200 mg daily divided on two occasions

Hyperthyroidism

The recommended dosage is 150-200 mg daily, divided in 3-4 doses. If needed, the dose can be increased.

Impaired Renal Function

The elimination rate is insignificantly affected by renal function, and dose adjustment is not needed in patients with impaired renal function.

Impaired Hepatic Function

Usually, Betatab 100mg Tablet is given in the same dose to patients suffering from liver cirrhosis as to patients with normal liver function. Only when there are signs of serious impairment of liver function (e.g., shunt-operated patients), a dose reduction should be considered.

Elderly

Dose adjustment is not needed.

Children

There is limited experience with Betatab 100mg Tablet treatment in children.

ROUTE OF ADMINISTRATION

Oral

CONTRAINDICATIONS

- Known hypersensitivity to metoprolol, related derivatives, any other β -blockers or to any of the excipients contained in the product.
- Second- or third-degree atrioventricular block.
- Uncontrolled heart failure.
- Clinically relevant sinus bradycardia (< 45-50 bpm).
- Sick sinus syndrome (unless a pacemaker is in situ).
- Prinzmetal's angina.
- Myocardial infarction complicated by significant bradycardia, first degree heart block, systolic hypotension (less than 100mmHg) and/or severe heart failure and cardiogenic shock.
- Severe peripheral arterial disease.
- Asthma and history of bronchospasm.
- Untreated pheochromocytoma.
- Metabolic acidosis.
- Concomitant intravenous administration of calcium blockers of the type verapamil or diltiazem or other antiarrhythmics (such as disopyramide) is contraindicated (exception: intensive care unit).
- Hypotension.
- Diabetes if associated with frequent episodes of hypoglycaemia.
- Chronic obstructive pulmonary disease.

WARNINGS & PRECAUTIONS

Intravenous administration of the verapamil-type should not be given to patients treated with β -blockers. Metoprolol may aggravate the symptoms of peripheral arterial circulatory disorders. Severely

impaired renal function. Serious acute conditions with metabolic acidosis. Combination treatment with digitalis.

Betatab 100mg Tablet should not be given to patients with latent or manifested heart insufficiency without concomitant treatment. In patients with Prinzmetal's angina the frequency and the extent of angina attacks may increase due to alpha-receptor mediated contraction of the coronary vessels. For this reason, non-selective beta-blockers must not be used in these patients. Beta₁-selective receptor blockers should be used with caution. During treatment with metoprolol the risk for interfering with carbohydrate metabolism or masking hypoglycemia is less than with non-selective beta-blockers. Very rarely, a pre-existing AV conduction disorder of moderate degree may become aggravated (possibly leading to AV block).

Although cardio-selective beta blockers may have less effect on lung function than non-selective beta blockers these should be avoided in patients with reversible obstructive airways disease unless there are compelling clinical reasons for their use. Although metoprolol has proved safe in a large number of asthmatic patients, it is advisable to exercise care in the treatment of patients with chronic obstructive pulmonary disease. Therapy with a beta₂-stimulant may become necessary or current therapy require adjustment. Therefore, non-selective beta blockers should not be used for these patients, and beta₁-selective blockers only with the utmost care.

Treatment with beta-blockers may aggravate the treatment of an anaphylactic reaction. Adrenaline treatment in normal dose does not always give the expected therapeutic effect. If Betatab 100mg Tablet is given to a patient with pheochromocytoma, treatment with an alpha-blocker should be considered.

Withdrawal of Betatab 100mg Tablet should, when possible, be done gradually over 2 weeks. The dose should be reduced gradually until a final dose of 25 mg is reached. During this period especially, patients with known ischemic heart disease should be kept under close surveillance. The risk for coronary events, including sudden death, may increase during withdrawal of beta-blockers.

Prior to surgery the anesthetist should be informed that the patient is receiving Metoprolol Tartrate (Betatab 100mg Tablet). It is not recommended to stop β -blocker treatment in patients undergoing surgery. Acute initiation of high-dose metoprolol to patients undergoing non-cardiac surgery should be avoided, since it has been associated with bradycardia, hypotension and stroke including fatal outcome in patients with cardiovascular risk factors.

The tablet contains lactose. Patients with any of the following rare hereditary conditions should not use this medicine: galactose intolerance, total lactase deficiency or glucose-galactose malabsorption.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

As dizziness and fatigue may occur in Betatab treatment, this should be considered when strict attention is required when driving or operating machines.

INTERACTIONS WITH OTHER MEDICAMENTS

- Anaesthetic drugs may attenuate reflex tachycardia and increase the risk of hypotension. Metoprolol therapy should be reported to the anaesthetist before the administration of a general anaesthetic. If possible, withdrawal of metoprolol should be completed at least 48 hours before anaesthesia. However, for some patients undergoing elective surgery, it may be desirable to employ a beta-blocker as premedication. By shielding the heart against the effect of stress,

metoprolol may prevent excessive sympathetic stimulation which is liable to provoke such cardiac disturbance as arrhythmias or acute coronary insufficiency during induction and intubation. Anaesthetic agents causing myocardial depression, such as cyclopropane and trichlorethylene, are best avoided. In a patient under beta-blockade an anaesthetic with as little negative inotropic activity as possible (halothane/nitrous oxide) should be selected.

- It may be necessary to adjust the dose of the hypoglycaemic agent in labile or insulin-dependent diabetes. Beta-adrenergic blockade may prevent the appearance of signs of hypo-glycaemia (tachycardia).
- Like all beta-blockers, metoprolol should not be given in combination with calcium channel blockers i.e. verapamil and to a lesser extent diltiazem since this may cause bradycardia, hypotension, heart failure and asystole and may increase auriculoventricular conduction time. However, combinations of antihypertensive drugs may often be used with benefit to improve control of hypertension. Calcium blockers of the verapamil type should not be administered intravenously to patients receiving beta blockers.
- Care should also be taken when beta-blockers are given in combination with sympathetic ganglion blocking agents, other beta blockers or MAO inhibitors. Concomitant administration of tricyclic antidepressants, barbiturates and phenothiazines as well as other antihypertensive agents may increase the blood pressure lowering effect.
- Calcium channel blockers (such as dihydropyridine derivatives e.g. nifedipine) should not be given in combination with metoprolol because of the increased risk of hypotension and heart failure. In patients with latent cardiac insufficiency, treatment with beta-blocking agents may lead to cardiac failure. Beta-blockers used in conjunction with clonidine increase the risk of "rebound hypertension". If combination treatment with clonidine is to be discontinued, metoprolol should be withdrawn several days before clonidine.
- The effects of metoprolol and other antihypertensive drugs on blood pressure are usually additive, and care should be taken to avoid hypotension.
- NSAIDs (especially indometacin) may reduce the antihypertensive effects of beta-blockers possibly by inhibiting renal prostaglandin synthesis and/or causing sodium and fluid retention.
- Digitalis Glycosides and/or diuretics should be considered for patients with a previous history of heart failure or in patients known to have a poor cardiac reserve. Digitalis glycosides in association with beta-blockers may increase in auriculo-ventricular conduction time.
- The administration of adrenaline (epinephrine) or noradrenaline (norepinephrine) to patients undergoing beta-blockade can result in an increase in blood pressure and bradycardia, although this is less likely to occur with beta₁-selective drugs. As beta-blockers may affect the peripheral circulation, care should be exercised when drugs with similar activity e.g. ergotamine are given concurrently. Concurrent use of moxislyte may result in possible severe postural hypotension.
- The effect of adrenaline (epinephrine) in the treatment of anaphylactic reactions may be weakened in patients receiving beta blockers.
- Metoprolol will antagonise the beta₁-effects of sympathomimetic agents but should have little influence on the bronchodilator effects of beta₂-agonists at normal therapeutic doses.
- Enzyme inducing agents (e.g. rifampicin) may reduce plasma concentrations of metoprolol, whereas enzyme inhibitors (e.g. cimetidine, hydralazine and alcohol), selective serotonin reuptake inhibitors (SSRIs) as paroxetine, fluoxetine and sertraline, diphenhydramine, hydroxychloroquine, celecoxib, terbinafine may

increase plasma concentrations of hepatically metabolized beta blockers.

- As with all beta-blockers particular caution is called for when metoprolol is administered together with prazosin for the first time as the co-administration of metoprolol and prazosin may produce a first dose hypotensive effect.
- Class 1 antiarrhythmic drugs, e.g. disopyramide, quinidine and amiodarone may have potentiating effects on atrial conduction time and induce negative inotropic effect. Concurrent use of propafenone may result in significant increases in plasma concentrations and half-life of metoprolol. Plasma propafenone concentrations are unaffected. Dosage reduction of metoprolol may be necessary.
- During concomitant ingestion of alcohol and metoprolol the concentration of blood alcohol may reach higher levels and may decrease more slowly. The concomitant ingestion of alcohol may enhance hypotensive effects.
- Metoprolol may impair the elimination of lidocaine.
- Prostaglandin synthetase inhibiting drugs may decrease the hypotensive effects of beta-blockers.
- Concurrent use of oestrogens may decrease the antihypertensive effect of beta-blockers because oestrogen induced fluid retention may lead to increased blood pressure.
- Concurrent use of xanthines, especially aminophylline or theophylline, may result in mutual inhibition of therapeutic effects.
- Xanthine clearance may also be decreased especially in patients with increased theophylline clearance induced by smoking.
- Concurrent use requires careful monitoring.
- Concurrent use of aldesleukin and alprostadil may result in an enhanced hypotensive effect.
- There is an increased risk of bradycardia following concomitant use of mefloquine with metoprolol.
- Concomitant use with anxiolytics and hypnotics may result in an enhanced hypotensive effect.
- Concomitant use with corticosteroids may result in antagonism of the hypotensive effect.

PREGNANCY AND LACTATION

Use in Pregnancy: Metoprolol Tartrate (Betatab) should not be given during pregnancy and lactation unless its use is considered essential. β-blockers may cause bradycardia, in the fetus and in the newborn and breastfed infant. This should be considered if these drugs are prescribed in the last trimester and in association with delivery. Betatab should gradually be withdrawn 48-72 hours before planned childbirth. If this is not possible the newborn infant should be supervised during 48-72 hours postpartum for signs and symptoms of *betablockade* (eg, heart- and lung complications).

Use in Lactation: Metoprolol is concentrated in human breast milk in a quantity that corresponds to approximately three times the quantity found in the plasma of the mother. The risk for harmful reactions with respect to the breast-feeding child seems to be low at therapeutic doses of the medicine. The breast-feeding child should however be observed regarding signs of betablockade.

SIDE EFFECTS

The following are the side effects of Betatab 100mg Tablet:

General disorders

- Fatigue and dysgeusia (taste disturbances).

Blood and lymphatic system disorders

- Thrombocytopenia and agranulocytosis.

Psychiatric disorders

- Depression, nightmares, nervousness, anxiety, impotence, hallucinations, personality disorder, amnesia or memory impairment.

Nervous system disorders

-Dizziness, headache, alertness decreased, somnolence or insomnia and paraesthesia.

Eye disorders

-Visual disturbance (e.g. blurred vision, dry eyes and/or eye irritation).

Ear and labyrinth disorders

-Tinnitus, and, in doses exceeding those recommended, "hearing disorders (eg. hypoacusis or deafness).

Cardiac disorders

-Bradycardia, heart failure, cardiac arrhythmias, palpitation, cardiac conduction disorders, precordial pain and increase in existing intermittent claudication.

Vascular disorders

-Orthostatic hypotension (occasionally with syncope), oedema, Raynaud's phenomenon and Gangrene in patients with pre-existing severe peripheral circulatory disorders.

Respiratory, thoracic and mediastinal disorders

-Exertional dyspnea, bronchospasm (which may occur in patients without a history of obstructive lung disease) and rhinitis.

Gastrointestinal disorders

-Nausea and vomiting, abdominal pain, diarrhoea or constipation and dry mouth.

Hepatobiliary disorders

-Hepatitis

Skin and subcutaneous tissue disorders

-Skin rash (in the form of urticaria, psoriasiform and dystrophic skin lesions), photosensitivity, hyperhidrosis, alopecia, worsening of psoriasis and occurrence of antinuclear antibodies (not associated with SLE).

Musculoskeletal and connective tissue disorders

-Muscle cramps and arthritis.

Reproductive system and breast disorders

-Disturbances of libido and potency.

SYMPTOMS & TREATMENT OF OVERDOSE***Symptoms:***

Poisoning due to an overdose of metoprolol may lead to severe hypotension, sinus bradycardia, atrioventricular block, heart failure, cardiogenic shock, cardiac arrest, bronchospasm, impairment of consciousness, coma, nausea, vomiting, cyanosis, hypoglycaemia and, occasionally, hyperkalemia. The first manifestations usually appear 20 minutes to two hours after drug ingestion.

After ingestion of an overdose or in case of hypersensitivity, the patient should be kept under close supervision and be treated in an intensive-care ward. Absorption of any drug material still present in the gastrointestinal tract can be prevented by induction of vomiting, gastric lavage, administration of activated charcoal and a laxative. Artificial respiration may be required.

Treatment:

Bradycardia or extensive vagal reactions should be treated by administering atropine or methylatropine. Hypotension and shock should be treated with plasma/plasma substitutes and, if necessary, catecholamines. The beta-blocking effect can be counteracted by slow

intravenous administration of isoprenaline hydrochloride or dobutamine.

In refractory cases isoprenaline can be combined with dopamine. Administration of calcium ions, or the use of a cardiac pacemaker may also be considered. In patients intoxicated with hydrophilic beta-blocking agents haemodialysis or haemoperfusion may be considered.

INSTRUCTIONS FOR USE

Method of Administration: For oral administration.

STORAGE

Store below 30°C.

SHELF LIFE

Product should not be used beyond the expiry date imprinted on the product packaging.

PACK SIZE(S)

Blister Pack of 10's

a) 10 strips of 10 tablets

b) 50 strips of 10 tablets

REGISTRATION NO

MAL21056021AZ

For further information, please consult your doctor or your pharmacist.

**CONTROLLED MEDICINE
UBAT TERKAWAL****KEEP MEDICINES OUT OF REACH OF CHILDREN
JAUHI UBAT-UBATAN DARI KANAK-KANAK**

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